A negative answer to the scaled-power order-continuity question

Literature-implied answer for arXiv:1208.3498

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Status. `literature_implied_answer (negative answer)`. Hadwin–Kitover–Orhon's construction directly answers the question and also yields a band-irreducible version. This is an identification of known ingredients, not a new solution claim.

1 Source question

For a positive operator T, Gao and Troitsky define $\mathbb{R}_+T$ to be the smallest norm-closed semigroup containing T and closed under multiplication by nonnegative scalars. Thus it contains every positive scalar multiple of a power of T and every norm limit

$$\lim_j b_j T^{n_j} \qquad (b_j \geq 0,\ n_j \nearrow \infty).$$

In Section 8 of [1], PDF page 23, they ask whether

$$T \text{ positive and order-continuous} \quad \Longrightarrow \quad \mathbb{R}_+T \text{ consists of order-continuous operators.}$$

2 Direct literature implication

Example 11 of Hadwin, Kitover, and Orhon [2] constructs a Banach lattice X and a positive order-continuous operator A on X. Their Remark 15 (supporting PDF, p. 15) records that

$$A^n \longrightarrow P \quad \text{in operator norm,}$$

where P is a positive spectral projection that is not order-continuous (not even σ-order-continuous). Taking $b_j = 1$ and $n_j = j$ in the definition above gives $P \in \mathbb{R}_+A$. Hence the answer to the source question is negative.

The supporting paper does not explicitly identify the later wording in [1]; the implication is direct but agent-identified. This is why the result is classified as a literature-implied answer.

3 The band-irreducible context also fails

The source question occurs inside a section on band-irreducible semigroups. The same Example 11 yields the following stronger statement.

Proposition 1. *There are a Banach lattice X and a positive, order-continuous, band-irreducible operator B such that B^n converges in operator norm to a positive projection that is not order-continuous. In particular, $\mathbb{R}_+B$ does not consist of order-continuous operators.*

Proof. We use the notation and properties proved in Example 11 of [2]. The construction gives a Banach lattice

$$X = J \oplus \operatorname{span}\{\mathbf{1}\}, \qquad J = \{x \in X : x(0) = 0\},$$

and a positive order-continuous lattice homomorphism A satisfying $A\mathbf{1} = \mathbf{1}$, $A(J) \subseteq J$, and

$$\|(A|_J)^k\| \le \frac{1}{k!}. \tag{1}$$

It also constructs a positive order-continuous band-irreducible operator T and lets L be multiplication by the coordinate function t. The paper proves that, for every sufficiently small $\varepsilon > 0$,

$$B_\varepsilon = A + \varepsilon LT$$

is positive, order-continuous, and band irreducible.

Evaluation at zero is fixed by B_ε: indeed $(Ax)(0) = x(0)$ and $(LTx)(0) = 0$. Consequently J is invariant and the operator induced on X/J is the identity. Put

$$C_\varepsilon = B_\varepsilon|_J = A|_J + \varepsilon(LT)|_J.$$

By (1), $r(A|_J) = 0$. Upper semicontinuity of the spectral radius therefore permits us to choose $\varepsilon > 0$ so small that $r(C_\varepsilon) < 1$, while retaining all the preceding properties.

Relative to $X = J \oplus \operatorname{span}\{\mathbf{1}\}$, write

$$B_\varepsilon = \begin{pmatrix} C_\varepsilon & d \\ 0 & 1 \end{pmatrix} \qquad (d \in J).$$

Since $r(C_\varepsilon) < 1$, we have $C_\varepsilon^n \to 0$ and $\sum_{k=0}^{n-1} C_\varepsilon^k d \to (I - C_\varepsilon)^{-1} d$. Hence

$$B_\varepsilon^n \longrightarrow P = \begin{pmatrix} 0 & (I - C_\varepsilon)^{-1} d \\ 0 & 1 \end{pmatrix} = u \otimes \delta_0,$$

where $\delta_0(x) = x(0)$ and $u = P\mathbf{1}$ satisfies $u(0) = 1$.

Remark 15 of [2] notes that $J = \ker \delta_0$ is an ideal but not a band; equivalently, δ_0 is not order-continuous. A nonzero positive rank-one operator $u \otimes \delta_0$ is order-continuous only if δ_0 is order-continuous. Therefore P is not order-continuous, although every B_ε^n is. Since $P \in \mathbb{R}_+ B_\varepsilon$, the stronger assertion follows. ☐

4 Scope

The proposition settles the precise closure implication negatively, even in the band-irreducible setting. It does not assert that all of the Perron–Frobenius conclusions sought in [1] fail under every weaker hypothesis; it only removes order continuity of the entire closed semigroup as an automatic consequence of order continuity of its generator.

References

[1] N. Gao and V. G. Troitsky, *Irreducible semigroups of positive operators on Banach lattices*, Linear Multilinear Algebra **62** (2014), 74–95; arXiv:1208.3498.

[2] D. W. Hadwin, A. K. Kitover, and M. Orhon, *Strong monotonicity of spectral radius of positive operators*, Houston J. Math. **41**, no. 2; arXiv:1205.5583v2.